Laws for Boolean Algebra

These are the rules we use to simplify logical expressions and design efficient circuits.

1. Identity Law

In the Boolean Algebra, we have identity elements for both AND(.) and OR(+) operations. The identity law states that in Boolean algebra, we have such variables that, on operating with the AND and OR operations we get the same result, i.e.

$$A + 0 = A$$

$$A.1 = A$$

2. Commutative Law

Binary variables in Boolean Algebra follow the commutative law. This law states that operating Boolean variables A and B is similar to operating Boolean variables B and A. That is,

$$A.\,B = B.\,A$$

$$A + B = B + A$$

3. Associative Law

Associative law states that the order of performing Boolean operator is illogical as their result is always the same. This can be understood as,

$$(A \,.\, B)\,.\,C = A \,.\,(B \,.\, C)$$

$$(A + B) + C = A + (B + C)$$

4. Distributive Law

Boolean Variables also follow the distributive law, and the expression for the Distributive law is given as:

$$A \,.\,(B + C) = (A \,.\, B) + (A \,.\, C)$$

5. Inversion Law

Inversion law is the unique law of Boolean algebra that states, the complement of the complement of any number is the number itself.

$$(A')' = A$$

6. AND Law

AND law of the Boolean algebra uses AND operator and the AND law is,

$$A \,.\, 0 = 0$$

$$A \,.\, 1 = A$$

$$A \,.\, A = A$$

7. OR Law

OR law of the Boolean algebra uses OR operator and the OR law is,

$$A + 0 = A$$

$$A + 1 = 1$$

$$A + A = A$$

8. Complement Law

The **Complement Law** states that a variable ORed with its complement is always 1, and a variable ANDed with its complement is always 0.

$$A + A' = 1$$

$$A \,.\, A' = 0$$

9. Domination Law

The **Domination Law** states that any variable ORed with 1 will always be 1, and any variable ANDed with 0 will always be 0.

$$A + 1 = 1$$

$$A \,.\, 0 = 0$$

10. Double Negation Law

The **Double Negation Law** states that the complement of the complement of a variable is the variable itself.

$$(A')' = A$$

The basic laws of Boolean Algebra are summarized in the table below:

Law	OR Form	AND Form
1. Identity Law	$P + 0 = P$	$P \cdot 1 = P$
2. Idempotent Law	$P + P = P$	$P \cdot P = P$
3. Commutative Law	$P + Q = Q + P$	$P \cdot Q = Q \cdot P$
4. Associative Law	$P + (Q + R) = (P + Q) + R$	$P \cdot (Q \cdot R) = (P \cdot Q) \cdot R$
5. Distributive Law	$P + (Q \cdot R) = (P + Q) \cdot (P + R)$	$P \cdot (Q + R) = (P \cdot Q) + (P \cdot R)$
6. Inversion Law	$(A')' = A$	$(A')' = A$

Law	OR Form	AND Form
7. De Morgan's Law	$(P + Q)' = P' \cdot Q'$	$(P \cdot Q)' = P' + Q'$
8. Complement Law	$P + P' = 1$	$P \cdot P' = 0$
9. Domination Law	$P + 1 = 1$	$P \cdot 0 = 0$
10. Double Negation Law	$(P')' = P$	$(P')' = P$
11. Absorption Law	$P + (P \cdot Q) = P$	$P \cdot (P + Q) = P$

De Morgan's Theorems

There are two basic theorems of great importance in Boolean Algebra, which are **De Morgan's First Law** and **De Morgan's Second Law.** These are also called De Morgan's Theorems. Now let's learn about both in detail.

De Morgan's First Law

De Morgan's Law states that the complement of the product (AND) of two Boolean variables (or expressions) is equal to the sum (OR) of the complement of each Boolean variable (or expression).

$$(P.Q)' = (P)' + (Q)'$$

The truth table for the same is given below:

P	Q	(P)'	(Q)'	(P.Q)'	(P)' + (Q)'
1	1	0	0	0	0
1	0	0	1	1	1
0	1	1	0	1	1
0	0	1	1	1	1

We can cee that truth values for (P.Q)' are equal to truth values for (P)' + (Q)', corresponding to the same input. Thus, De Morgan's first law is true.

De Morgan's Second Law

Statement: The Complement of the sum (OR) of two Boolean variables (or expressions) is equal to the product(AND) of the complement of each Boolean variable (or expression).
(P + Q)' = (P)'.(Q)'
Proof:
The truth table for the same is given below:

P	Q	(P)'	(Q)'	(P + Q)'	(P)'.(Q)'
1	1	0	0	0	0
1	0	0	1	0	0
0	1	1	0	0	0
0	0	1	1	1	1

We can see that truth values for (P + Q)' are equal to truth values for (P)'.(Q)', corresponding to the same input. Thus, De Morgan's second law is true.

Applications of Boolean Algebra

Boolean Algebra finds applications in many other fields of science related to digital logic design, computer science, telecommunications, etc. Some of its applications are:
Digital Logic Design:
Boolean Algebra acts as the backbone of digital logic design, being the most important element in the creation and analysis of digital circuits used in computers, smartphones, and all other electronic devices. It helps simplify the logic gates and circuits so that in the design of digital systems, they can be effectively designed and optimized.
Algorithm Design:

In computer science, Boolean Algebra is utilized in the design and study of algorithms, particularly in fields that require decision-making processes. It's vital in database query optimization, where Boolean logic is utilized to filter and obtain specific data based on circumstances.

Telecommunications:

Boolean Algebra finds application in the design and analysis of communication systems in telecommunication. More specifically, it is used in error detection and correction mechanisms. It is also used in the modulation and encoding of signals so that data is efficiently and accurately transmitted over networks.

Artificial Intelligence (AI):

Boolean Algebra is vital in AI, notably in the construction of decision-making algorithms and neural networks. It's used to model logical thinking and decision trees, which are crucial in machine learning and expert systems.

Electrical Engineering:

In electrical engineering, Boolean Algebra is employed to analyze and design switching circuits, which are important in the operation of electrical networks and systems. It aids in the optimization of these circuits, ensuring minimal energy loss and effective functioning.

Solved Examples on Boolean Algebra

Note: You can use True and False instead of 0,1 as well.

T = 1

F = 0

Question 1: Draw a Truth Table for P + P . Q = P

Solution:

The truth table for P + P . Q = P

P	Q	P . Q	P + P . Q
T	T	T	T
T	F	F	T
F	T	F	F

P	Q	P . Q	P + P . Q
F	F	F	F

In the truth table, we can see that the truth values for P + P.Q is exactly the same as P.

Question 2: Draw Truth Table for P . Q + P + Q
Solution:
The truth table for P . Q + P + Q

P	Q	P . Q	P . Q + P + Q
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

Some examples of simplification using the laws of boolean algebra:

1.F = A + A·B
= A(1 + B)
= A · 1 (1 + B = 1)
= A

2.F = A(A + B)
= A·A + A·B
= A + AB
= A(1+B)   [1+B=1]

= A.1

= A

3.F = (A + B)(A + C)

=A.A+A.B+A.C+BC //   Can be done directly

=A+AB+AC+BC   //

=A(1+B+C)+BC //
= A.1 + BC

=A+BC

4.F = A + B·C + A·C
= A + BC   (Absorption: A + AC = A)

5.F = A'B + AB + A'B'   // Try to find common things
= B(A' + A) + A'B' [A+A'=1]
= B·1 + A'B'
= B + A'B'   [(A+B)(A+C) = A+BC]   --> (B+A')(B+B')
= (B + A')(B + B')
= (B + A')(1)
= A' + B

(A+B).(A+C) = A+BC // Formulas can be used in reverse as well

// Sometimes, expressions might need to be expanded before simplification.

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6. $F = AB + A'C + BC$
$= AB + A'C + BC\cdot 1$
$= AB + A'C + BC(A + A')$ [Expanded further for simplification]
$= AB + A'C + ABC + A'BC$
$= AB + ABC + A'C + A'BC$
$= AB(1 + C) + A'C(1 + B)$
$= AB + A'C$
$= AB + A'C$

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7. $F = A'B' + A'B + AB$
$= A'(B' + B) + AB$
$= A' \cdot 1 + AB$
$= A' + AB$
$= (A' + A)(A' + B)$
$= 1\cdot(A' + B)$
$= A' + B$

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8. $F = A'BC + AB'C + ABC' + ABC$

$=C(A'B+AB) +ABC'+ABC$
$= C(A'B + AB') + AB(C' + C)$
$= C(A \oplus B) + AB \cdot 1$
$= C(A \oplus B) + AB$

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9. $F = A'B'C + A'BC + AB'C + ABC$
$= C(A'B' + A'B + AB' + AB)$
$= C[(A' + A)(B' + B)]$
$= C(1\cdot 1)$
$= C$

10.F = A'B + AB' + A'B' C
= (A'B + AB') + A'B'C
= A ⊕ B + A'B'C

[A'B'C can be ignored because if A=1, and if A=0, the expression becomes, which simplifies to by absorption. However, if we add to this, the entire expression becomes as well.]
= A ⊕ B

11.F = (A + B + C)(A + B' + C)(A' + B + C)
= (A + C + BB')(A' + B + C)   [Note: (A+C)(B)+(A+C)B'+(A+C)+0]
= (A + C + 0)(A' + B + C)
= (A + C)(A' + B + C)
= AA' + AB + AC + CA' + BC + C
= 0 + AB + AC + CA' + BC + C
= C + AB + AC + A'C
= C + AB
= C + AB

12.F = A'B'C + ABC' + A'BC + AB'C'
= C(A'B' + A'B) + C'(AB + AB')
= C(A') + C'(A)
= A'C + AC'
= A ⊕ C

13.F = A'BC' + AB'C' + AB'C + ABC
= C'(A'B + AB') + C(AB' + AB)
= C'(A ⊕ B) + C(A)
= AC + C'(A ⊕ B)

14.F = A'B' + AB' + ABC' + ABC
= B'(A' + A) + AB(C' + C)
= B'·1 + AB·1
= B' + AB

= B' + A B

15.F = (A + B)(A' + C)(B' + C')
= (AA' + AC + BA' + BC)(B' + C')
= (0 + AC + BA' + BC)(B' + C')
= (AC + BA' + BC)(B' + C')

= AB’C+0+0+0+A’BC’+0
= AB'C + A'BC'

16.F = A'B'C' + A'BC' + AB'C + ABC
= C'(A'B' + A'B) + C(AB' + AB)
= C' A' + C A
= A'C' + AC (X-NOR)